

Antenna Datasheet

Product OC: YECW058L1BM
Version: 1.0
Date: 2026-08-10
Status: Released

Product Name: 4G Terminal Mount External Dipole Antenna
Key Features:
Frequency Band: 790–960 MHz, 1710–2690 MHz
Dimensions: 223.18 mm × 74.09 mm × 76.50 mm
Efficiency: Up to 60.0 % (FS)
RoHS & REACH Compliant

Overview

YECW058L1BM is a 4G external antenna measuring 223.18 mm × 74.09 mm × 76.50 mm. This ultra-wide-band 4G antenna provides broad coverage from 790–960 MHz & 1710–2690 MHz whilst offering backward-compatibility to support 3G and 2G networks as well as LTE Cat-M and narrowband IoT (NB-IoT). The antenna is terminated with SMA Male connector. Ideal for applications where the antenna is required to be discrete, this low profile, terminal mount omni-directional antenna is easy to install with maximum durability assured thanks to its PC + ABS enclosure.

YECW0581BM is designed as dipole type to work with various GND plane sizes or in free space for ease of integration with a SMA Male connector to achieve the optimum position. This omni-directional antenna is ideally suited for Gateways & Routers, Smart Metering, Vending Machines, Industrial IoT, Smart Home, Connected Enterprise, offering great performance with its high gain and efficiency.

- **Typical applications include:**

✓ Gateways & Routers
✓ Smart Metering
✓ Vending Machines
✓ Industrial IoT
✓ Smart Home
✓ Connected Enterprise

Quectel provides comprehensive antenna design support such as simulation, testing and manufacturing for custom antenna solutions to meet your specific application needs. We have regional R & D centers to offer quick response to meet your requirements. Please contact our sales & FAEs if you have any requests.

Contents

Overview **1**
Contents **2**

1 Specification **3**
1.1. Electrical 3
1.2. Mechanical & Environmental 5

2 Drawing **6**

3 Detailed Performance **7**
3.1. S-Parameter Test 7
3.1.1. VSWR 7
3.1.2. Return Loss 8
3.2. Radiation Performance Test 9
3.2.1. Efficiency 9
3.2.2. Average Gain 10
3.2.3. Peak Gain 11
3.2.4. 3D & 2D Radiation Pattern 12
3.2.4.1. Test Condition: In Free Space 12
3.2.4.2. Test Condition: On 500 mm × 500 mm Metal Plane 16

4 Packaging **20**

Contact Us **22**
Legal Notices **23**
Revision History **25**

1 Specification

Test Condition: In Free Space & On 500 mm × 500 mm Metal Plane

1.1. Electrical

Electrical	
Frequency Range	790–960 MHz, 1710-2690 MHz
Impedance	50 Ω
Polarization	Linear
Radiation Pattern	Omni-directional

Electrical – Detail

Band / SPEC	Band	B71	B20	B5 /B8 /B26	n74 /n75 /n76	B1 /B2 /B3	B40	Wi-Fi 2G	B38 /B41	B42 /B48 /n77	n79	Wi-Fi 5G
	Freq. (MHz)	600–700	790–820	820–960	1420–1520	1700–2170	2300–2400	2400–2500	2500–2690	3300–4200	4400–5000	5150–5850
Max. VSWR	FS	-	1.1	1.8	-	1.6	1.9	1.9	1.8	-	-	-
	MP	-	1.7	2.2	-	1.6	1.9	1.9	2.0	-	-	-
Max. Return Loss (dB)	FS	-	-24.9	-11.1	-	-12.3	-10.0	-10.4	-10.8	-	-	-
	MP	-	-12.0	-8.3	-	-12.2	-10.3	-10.1	-9.7	-	-	-
AVG Eff. (%)	FS	-	59.6	54.7	-	46.3	36.3	35.1	36.3	-	-	-
	MP	-	56.4	48.4	-	46.0	37.9	34.9	35.1	-	-	-
AVG AVG Gain (dB)	FS	-	-2.3	-2.6	-	-3.3	-4.4	-4.5	-4.4	-	-	-
	MP	-	-2.5	-3.2	-	-3.4	-4.2	-4.6	-4.6	-	-	-
Max. Peak Gain (dBi)	FS	-	1.0	1.5	-	2.9	1.2	1.2	2.7	-	-	-
	MP	-	5.6	5.3	-	7.1	6.5	6.1	4.7	-	-	-

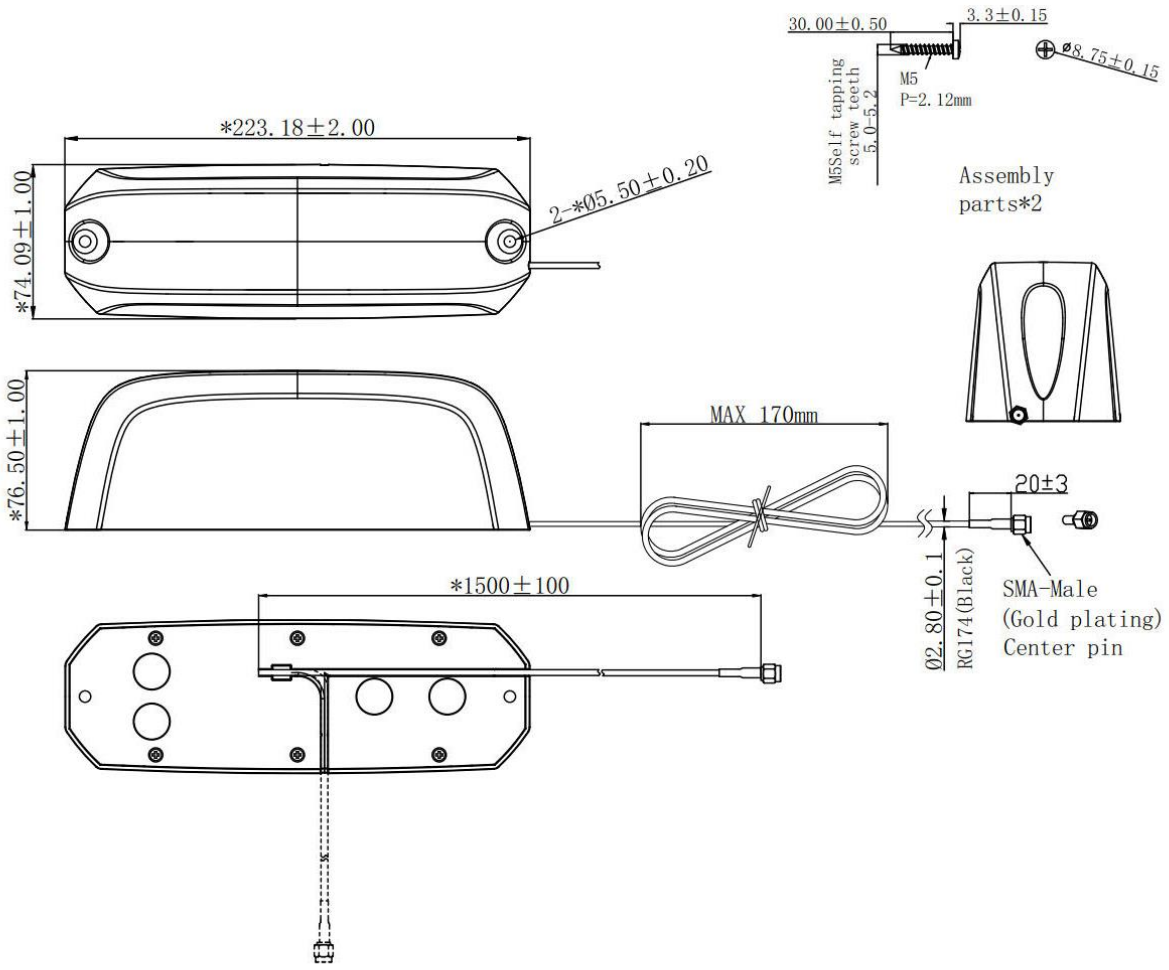
VSWR	FS	≤ 1.9
	MP	≤ 2.2
Return Loss	FS	≤ -10.0 dB
	MP	≤ -8.3 dB
Peak Gain	FS	≤ 2.9 dBi
	MP	≤ 7.1 dBi

- FS: In Free Space
- MP: On 500 mm × 500 mm Metal Plane

1.2. Mechanical & Environmental

Mechanical	
Antenna Dimensions	223.18 mm × 74.09 mm × 76.50 mm
Material & Color	PC + ABS & Black
Connector Type	SMA Male
Cable Type & Color & Length	RG174 & Black & 1500 mm
Mounting Type	Terminal
Weight	Typ. 219 g
Environmental	
Operation Temperature	-40 °C to +85 °C
Storage Temperature	-40 °C to +85 °C
RoHS & REACH Compliant	Yes

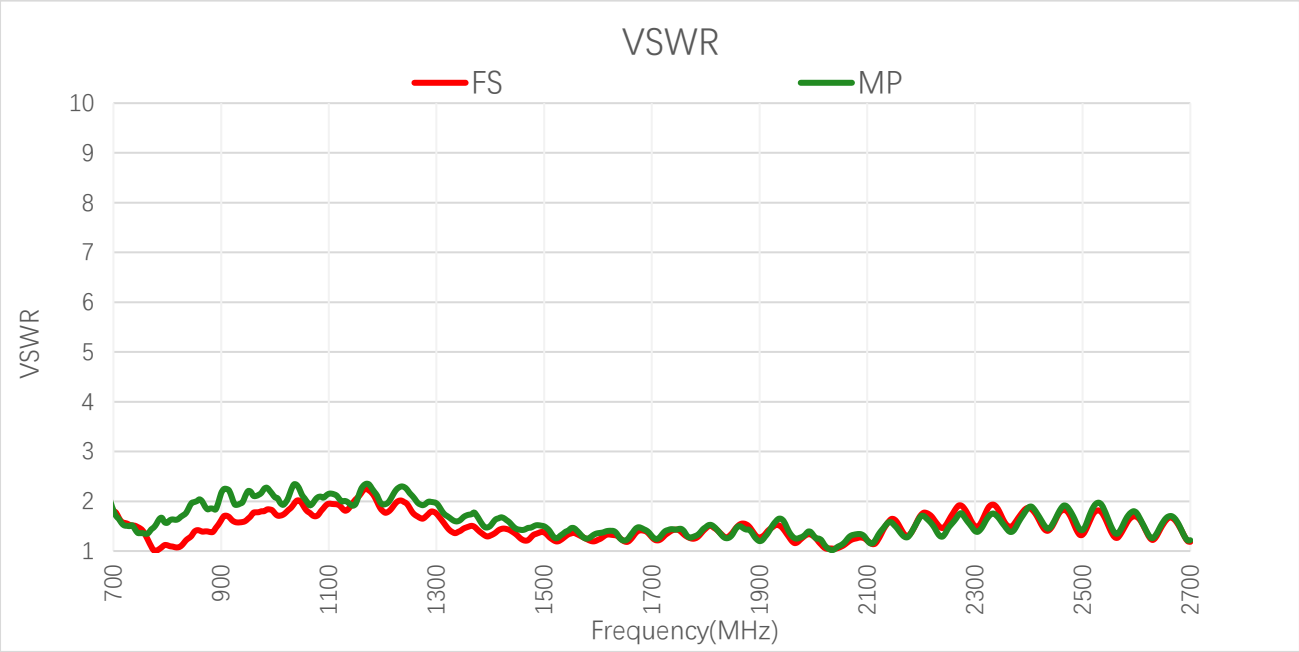
2 Drawing


30.00±0.50
3.3±0.15
ø8.75±0.15
M5
P=2.12mm
M5Self tapping screw teeth 5.0-5.2
Assembly parts*2
*223.18±2.00
2-*05.50±0.20
*74.09±1.00
*76.50±1.00
MAX 170mm
20±3
ø2.80±0.1
RG174(Black)
SMA-Male
(Gold plating)
Center pin
*1500±100

3 Detailed Performance

3.1. S-Parameter Test

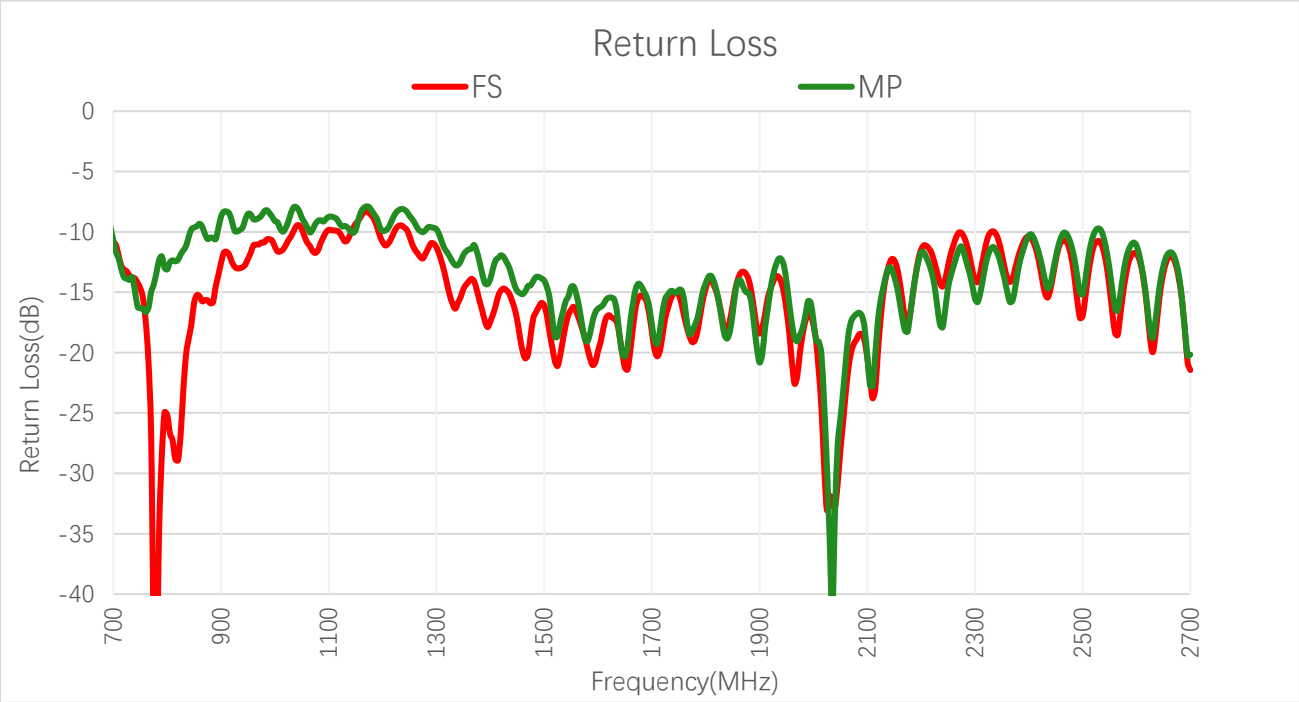
3.1.1. VSWR



VSWR

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
FS	-	-	-	1.2	1.6	1.8	-	1.2	1.4	1.5
MP	-	-	-	1.7	2.2	2.1	-	1.2	1.4	1.4
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
FS	1.4	1.6	1.7	1.6	1.7	1.3	-	-	-	-
MP	1.5	1.6	1.6	1.7	1.8	1.3	-	-	-	-

3.1.2. Return Loss

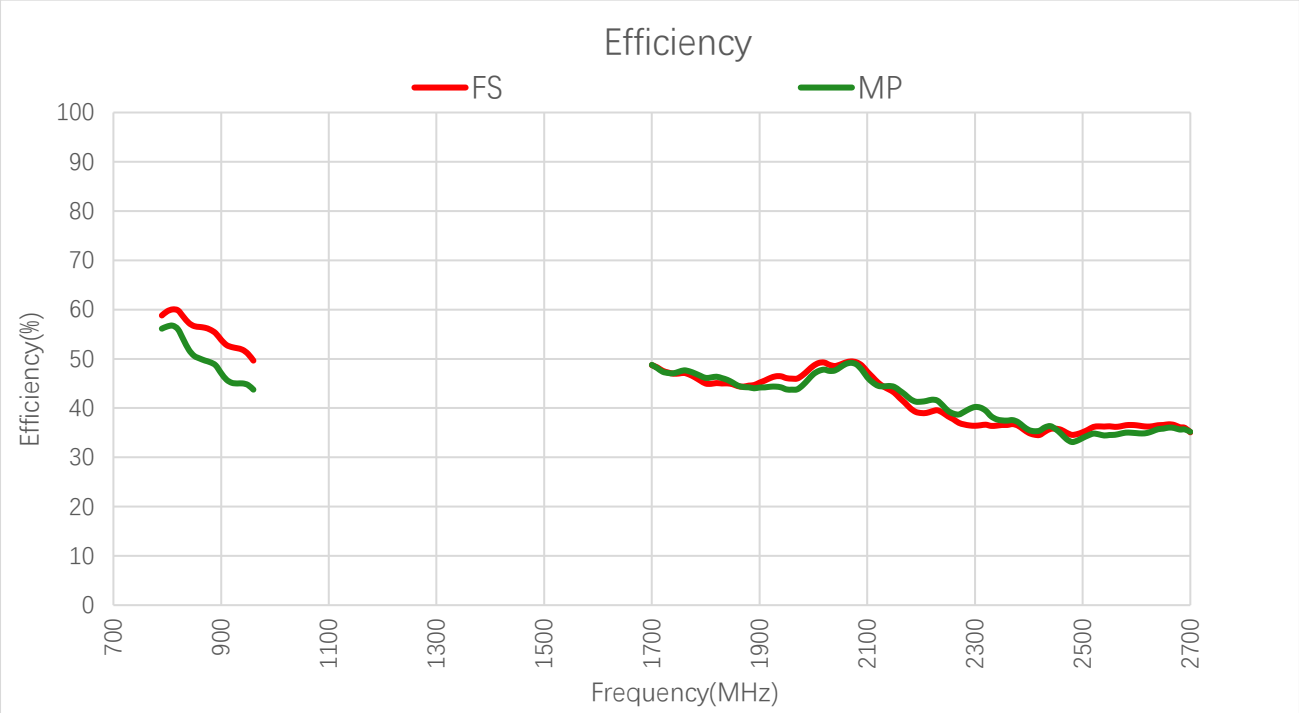


Return Loss (dB)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
FS	-	-	-	-23.0	-12.5	-11.1	-	-20.3	-15.2	-14.0
MP	-	-	-	-11.5	-8.7	-8.9	-	-19.3	-14.9	-15.1
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
FS	-16.5	-12.8	-11.9	-12.4	-11.8	-18.2	-	-	-	-
MP	-14.1	-13.0	-13.0	-12.1	-11.1	-18.2	-	-	-	-

3.2. Radiation Performance Test

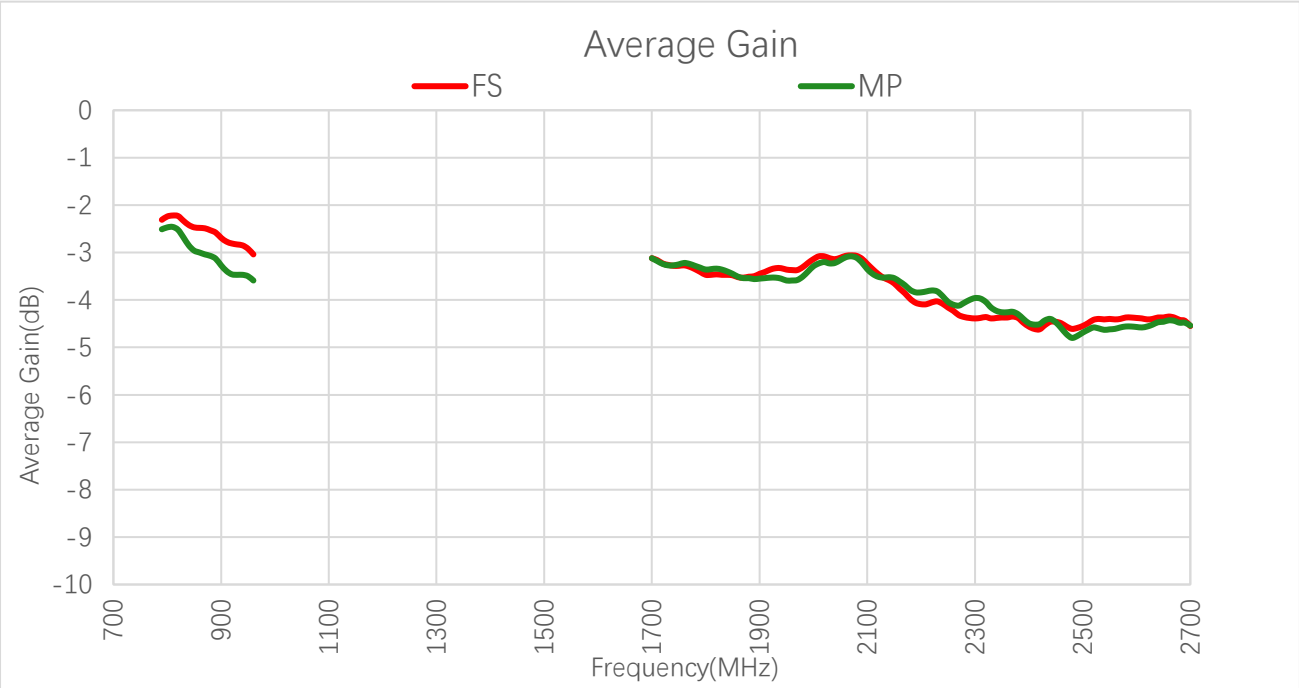
3.2.1. Efficiency



Efficiency (%)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
FS	-	-	-	58.5	53.9	49.6	-	48.3	47.0	44.6
MP	-	-	-	53.9	47.1	43.7	-	48.1	47.1	44.2
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
FS	46.1	43.8	36.6	35.8	36.5	36.0	-	-	-	-
MP	43.8	44.5	37.5	35.7	34.9	35.7	-	-	-	-

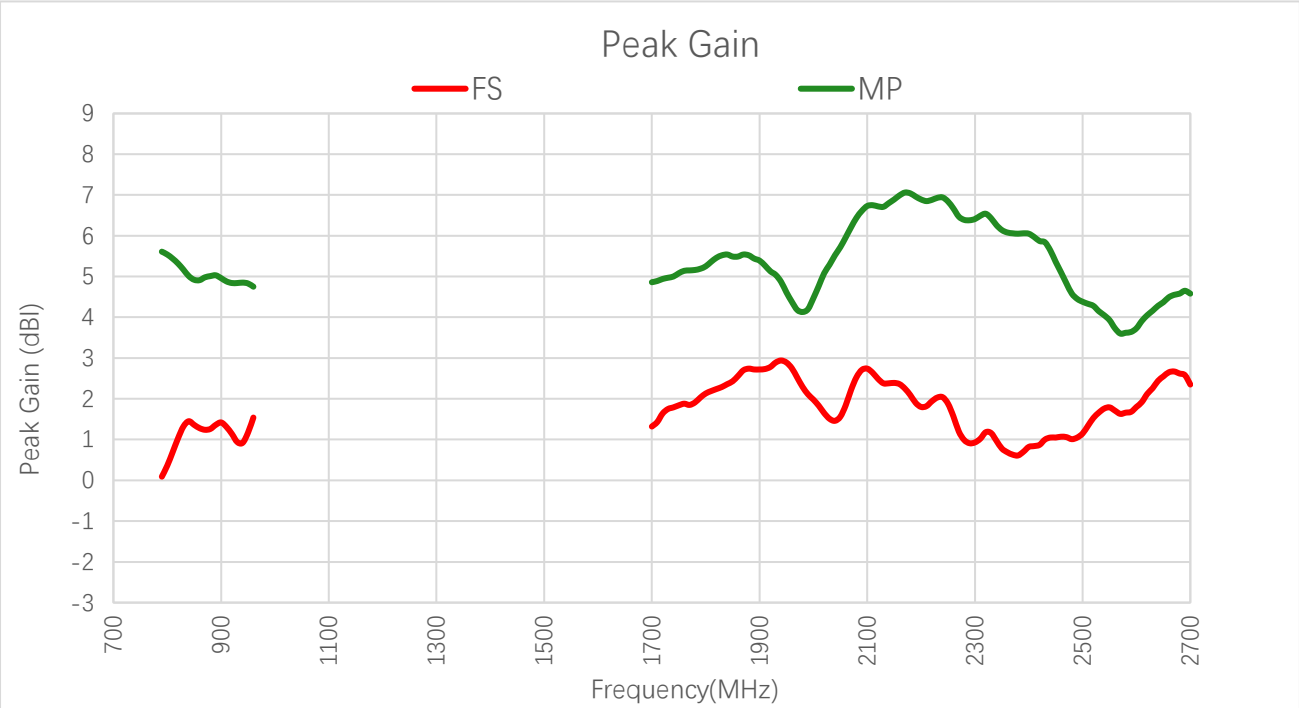
3.2.2. Average Gain



Average Gain (dB)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
FS	-	-	-	-2.3	-2.7	-3.0	-	-3.2	-3.3	-3.5
MP	-	-	-	-2.7	-3.3	-3.6	-	-3.2	-3.3	-3.5
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
FS	-3.4	-3.6	-4.4	-4.5	-4.4	-4.4	-	-	-	-
MP	-3.6	-3.5	-4.3	-4.5	-4.6	-4.5	-	-	-	-

3.2.3. Peak Gain



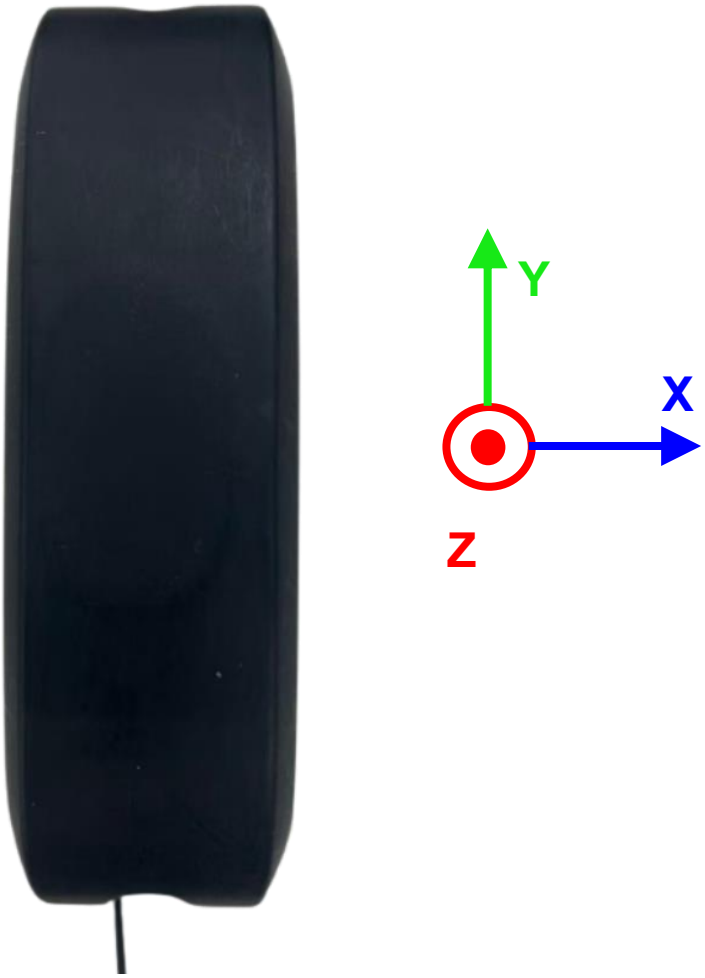
Peak Gain (dBi)

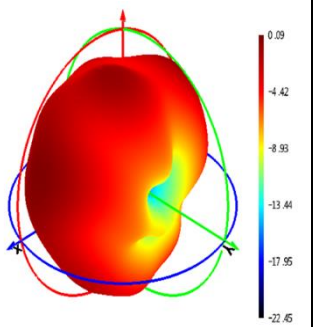
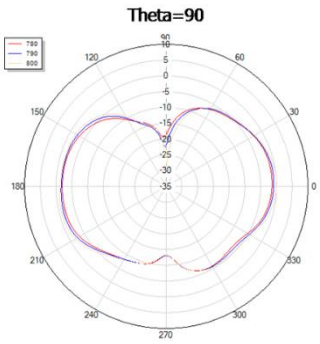
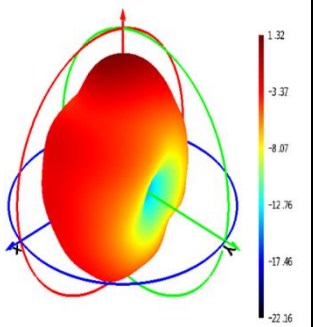
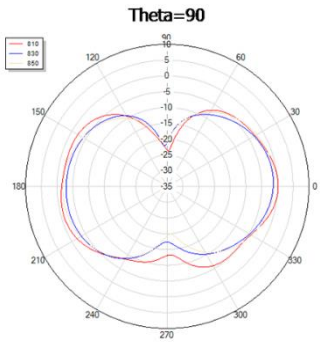
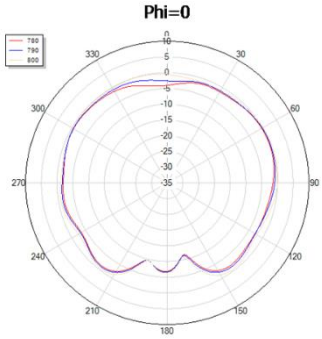
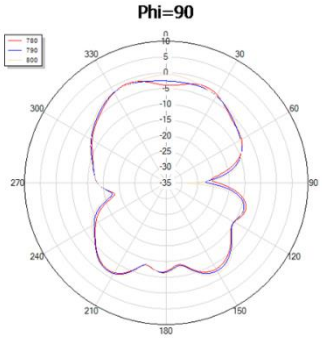
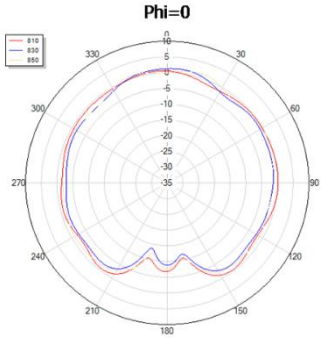
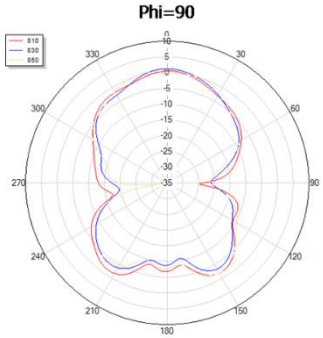
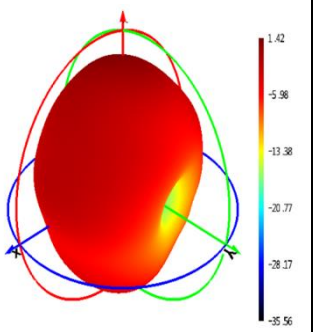
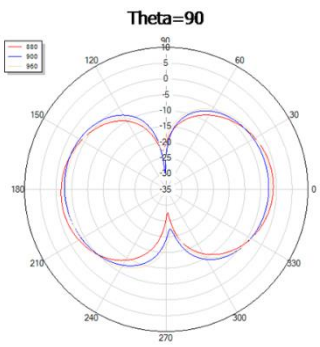
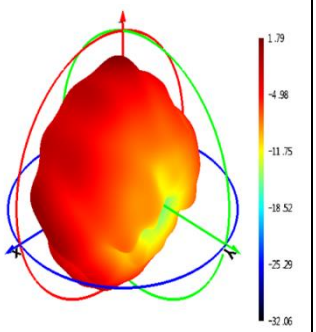
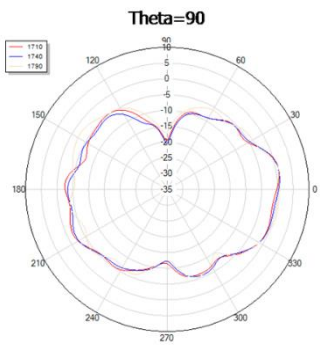
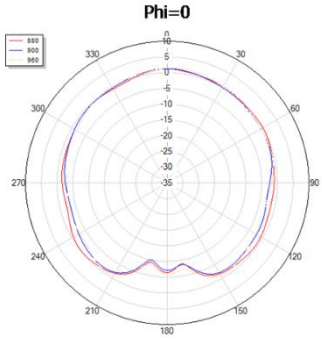
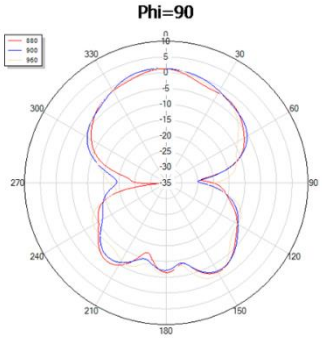
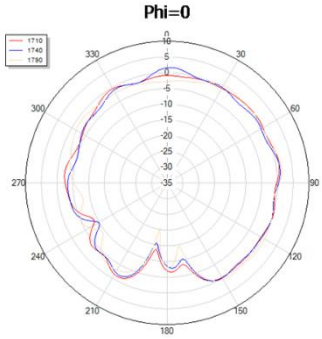
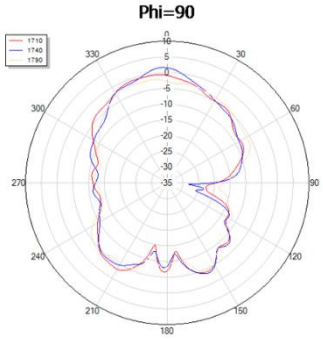
Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
FS	-	-	-	1.3	1.4	1.5	-	1.4	1.8	2.7
MP	-	-	-	5.2	5.0	4.8	-	4.9	5.0	5.5
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
FS	2.9	2.4	0.8	1.1	1.8	2.6	-	-	-	-
MP	4.6	6.8	6.1	5.4	3.7	4.7	-	-	-	-

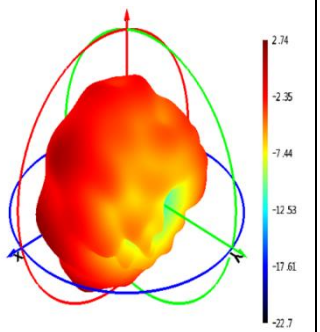
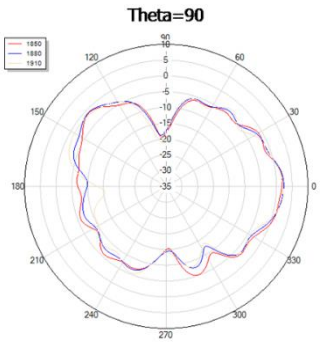
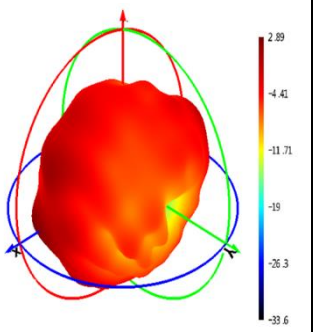
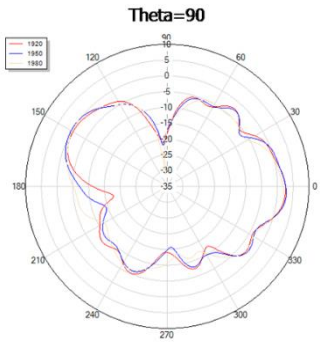
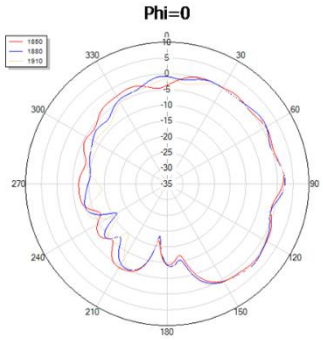
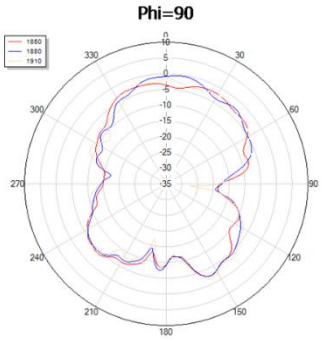
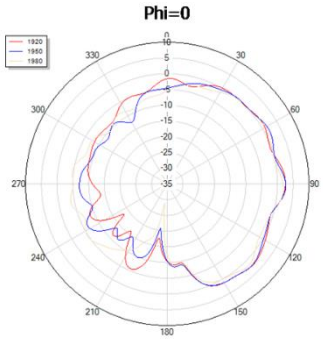
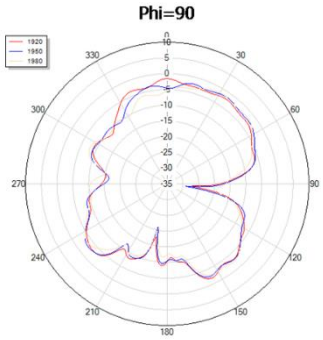
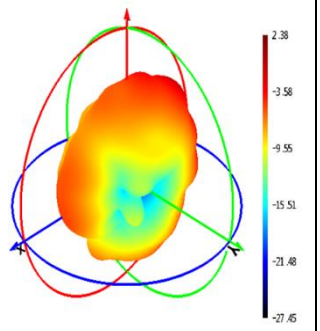
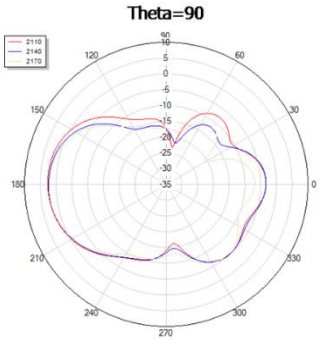
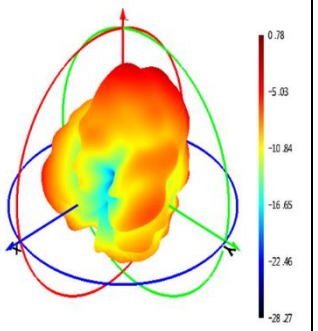
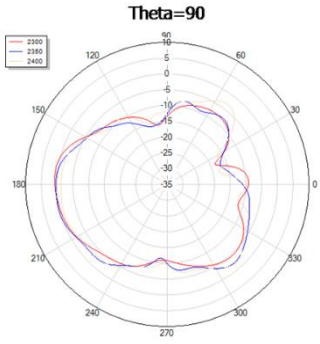
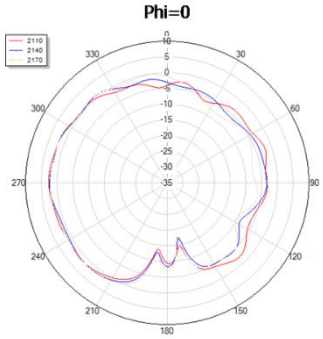
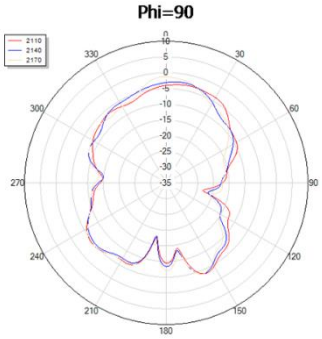
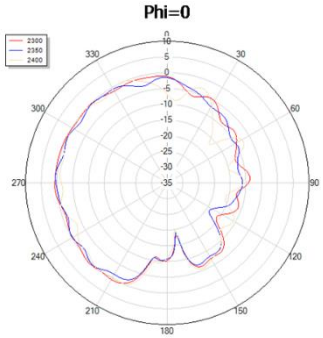
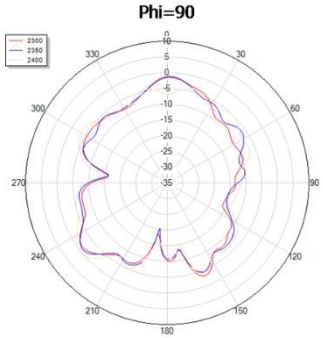
3.2.4. 3D & 2D Radiation Pattern

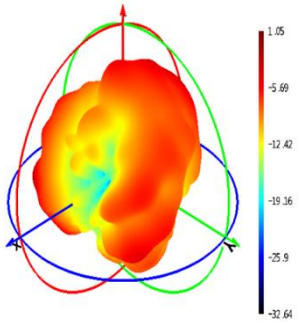
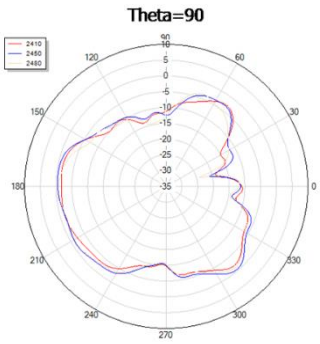
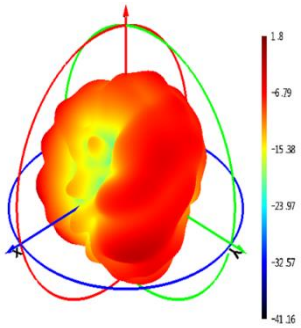
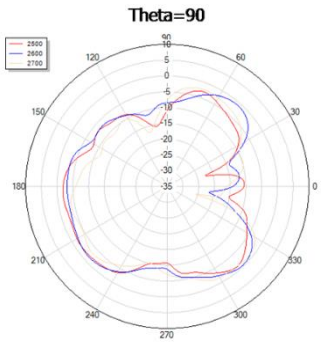
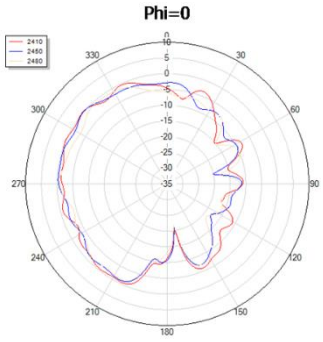
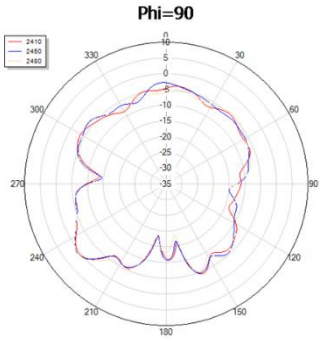
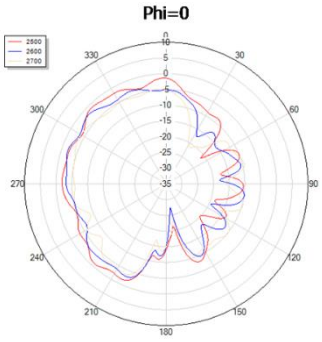
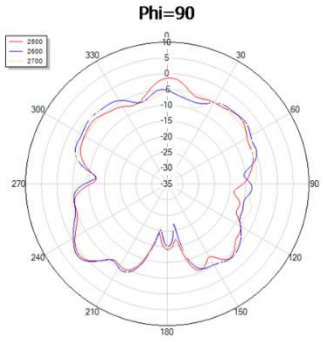
3.2.4.1. Test Condition: In Free Space

- Test Chamber: HF-S-1



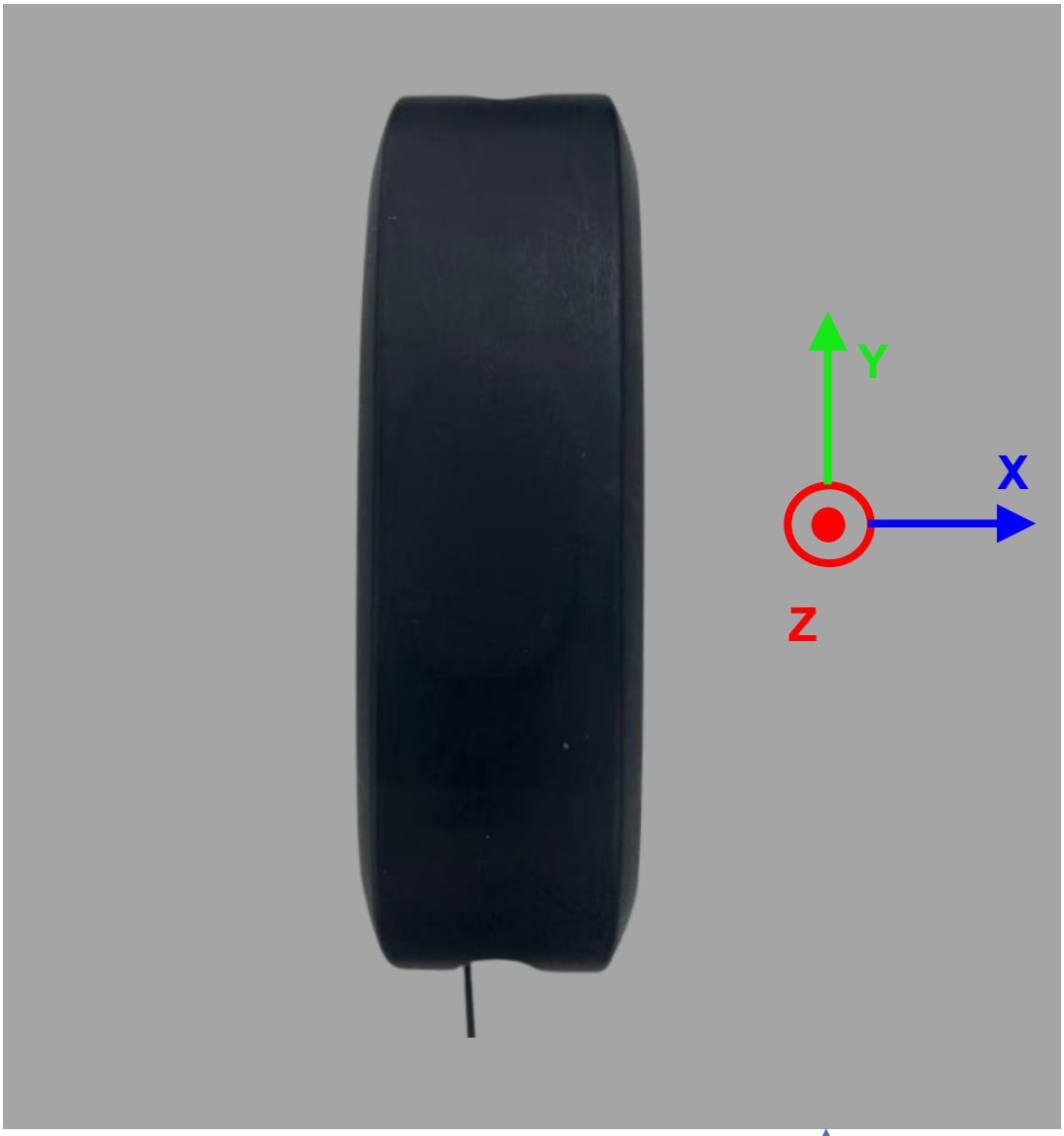
790 MHz		830 MHz	
			
			
900 MHz		**1740 MHz**	
			
			

1880 MHz		1950 MHz	
			
			
2140 MHz		**2350 MHz**	
			
			

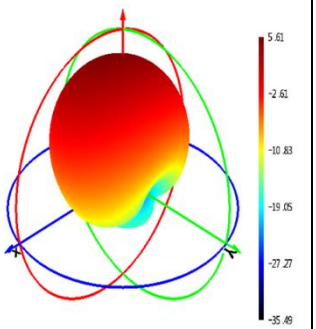
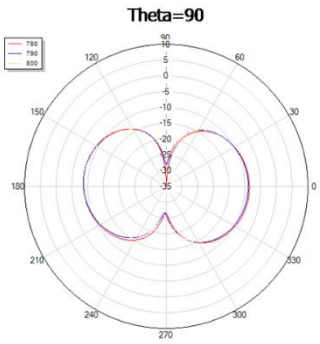
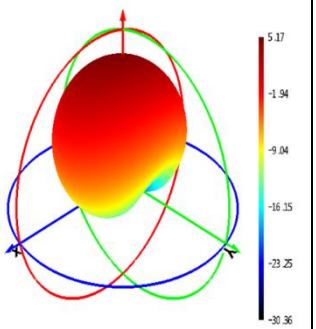
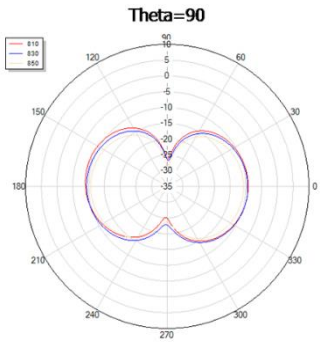
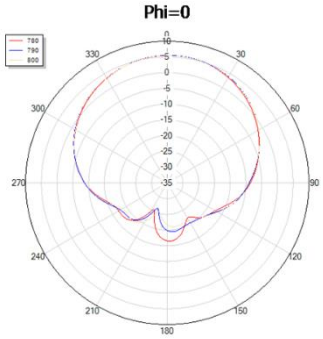
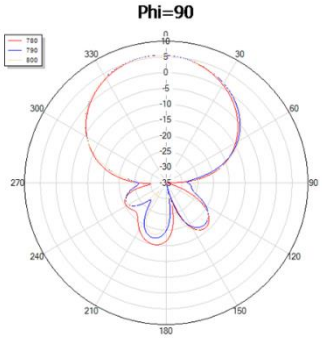
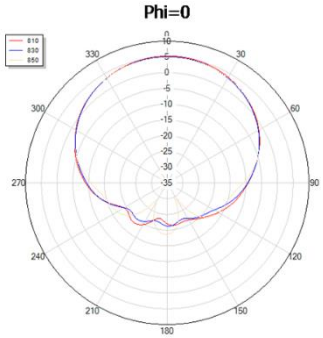
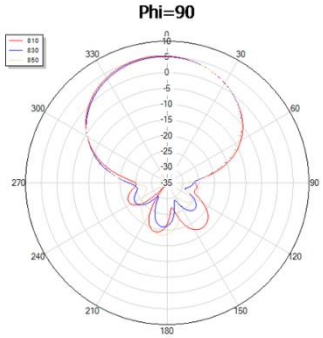
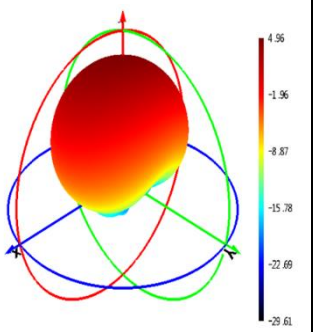
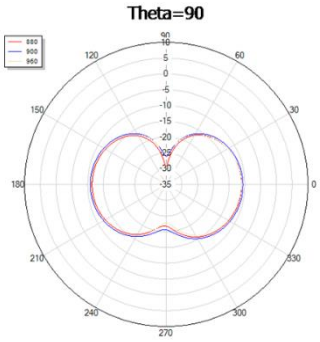
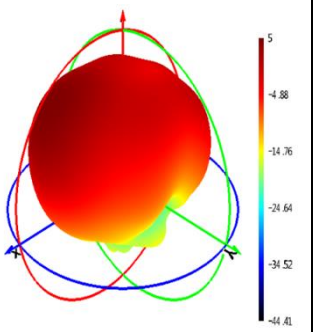
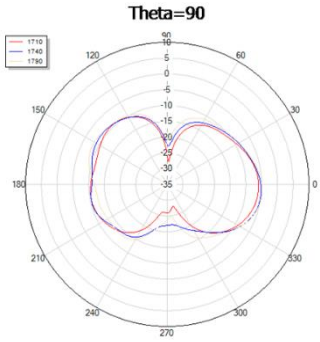
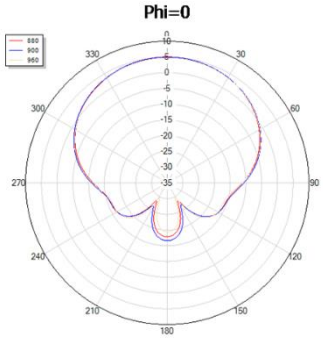
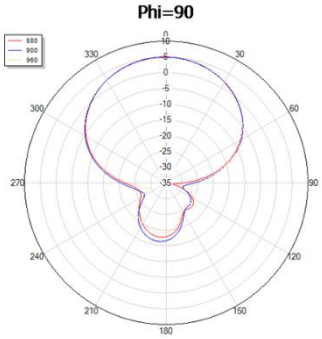
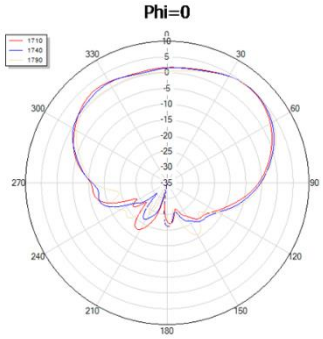
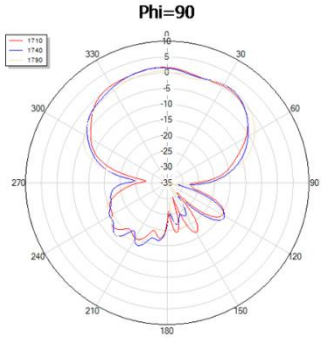
2450 MHz		2600 MHz	
			
			

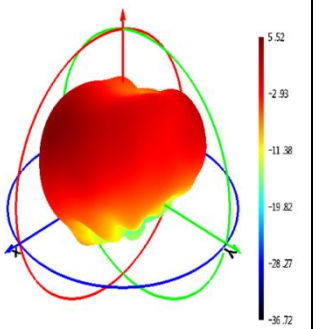
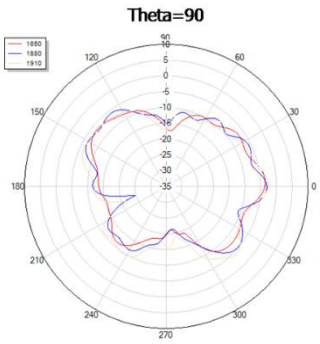
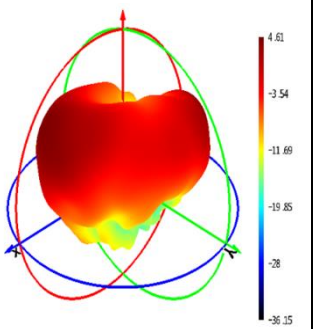
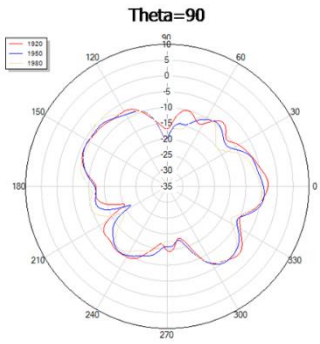
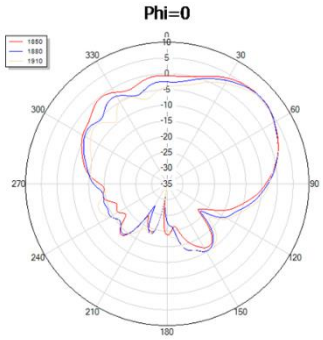
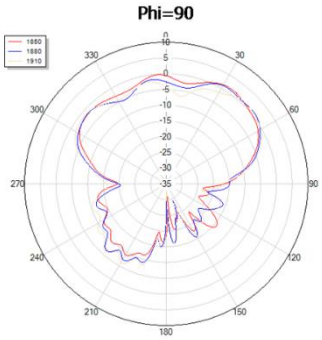
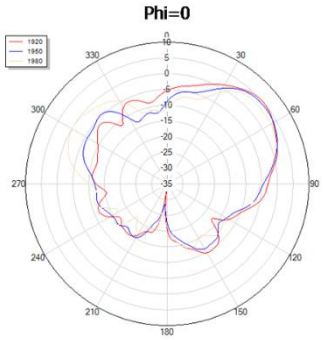
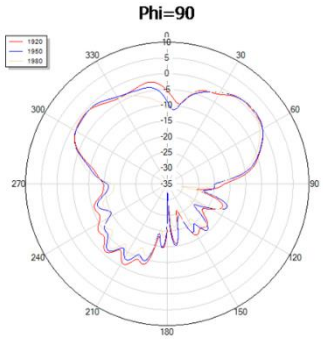
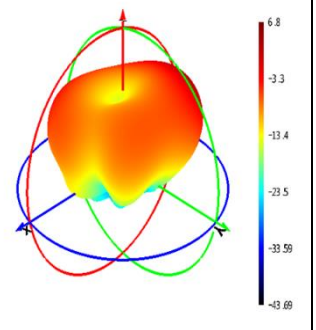
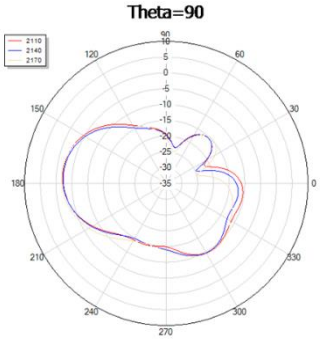
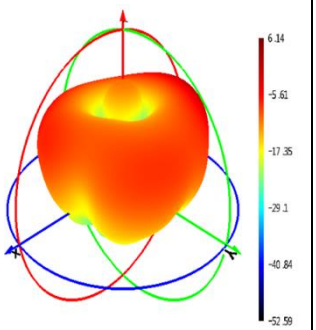
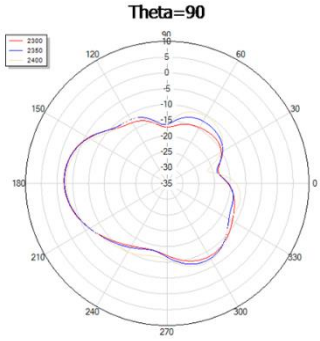
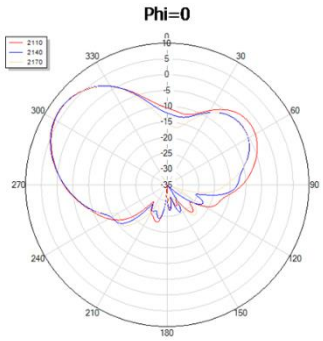
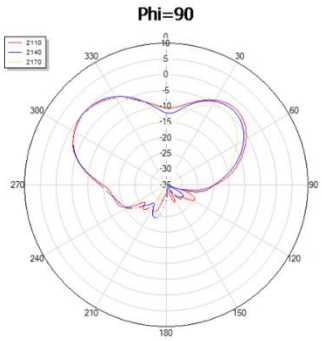
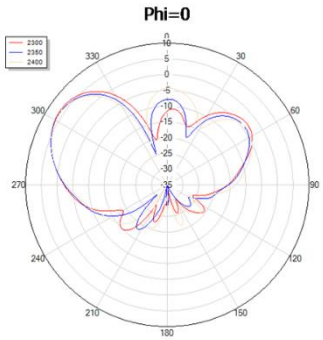
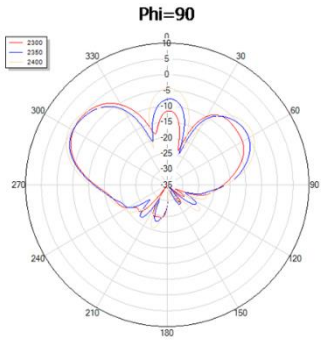
3.2.4.2. Test Condition: On 500 mm × 500 mm Metal Plane

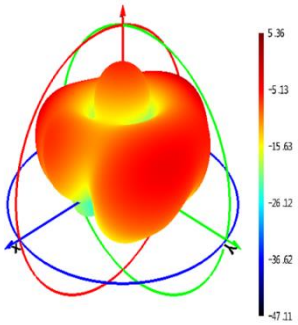
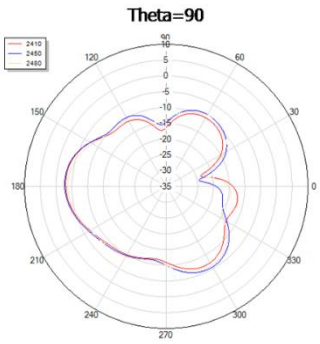
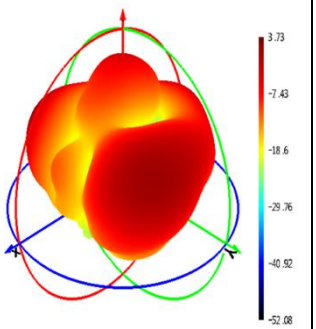
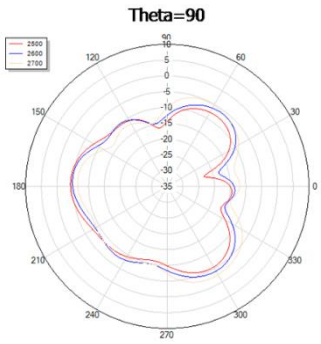
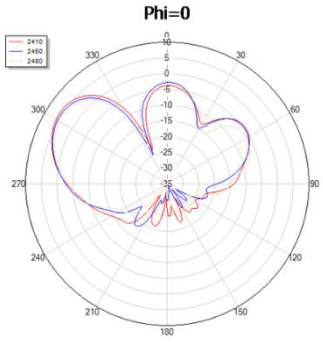
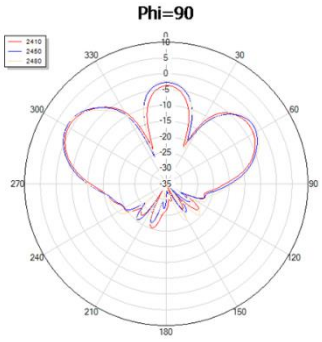
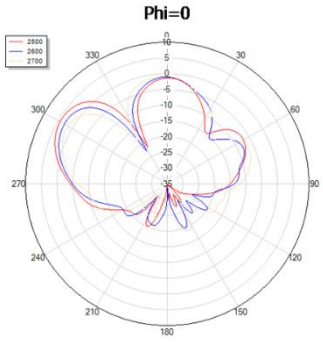
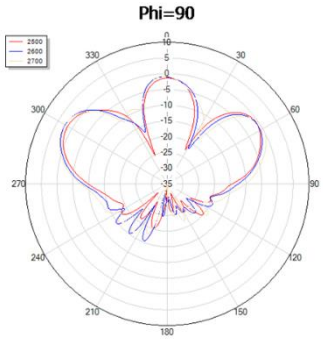
- Test Chamber: FS-S-1



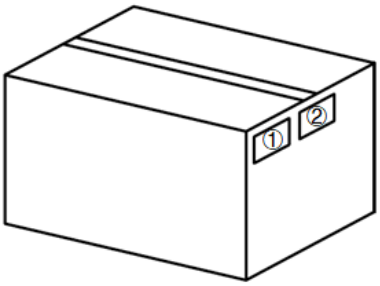
500 mm × 500 mm Metal Plane

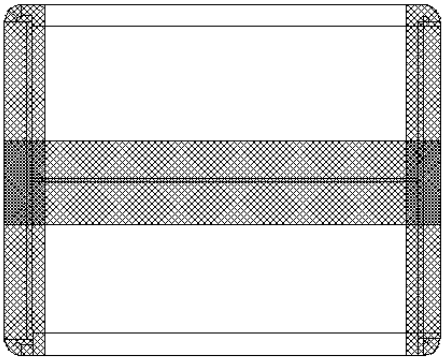
790 MHz		830 MHz	
			
			
900 MHz		1740 MHz	
			
			

1880 MHz		1950 MHz	
			
			
2140 MHz		**2350 MHz**	
			
			

2450 MHz		2600 MHz	
			
			

4 Packaging

Step	Packaging Picture / 2D Picture	Description
1		1 bubble bag contains 1 product.
2		1 inner box contains 2 products.
3		(12 Bubble Bags / Inner box) (24 Antennas / Carton Box) Estimated quantity Products that cannot fill the entire carton box are packed in a suitable size carton box. Carton Size: L × W × H = 500 × 420 × 330 mm
4		**Position for Attaching Labels** ① Carton Label ② Quality Label

5		**Sealing Cartons** H-shaped sealing cartons
Note	The initial packaging method described above is for reference only, and the final actual packaging method shall be subject to the actual shipping packaging.	

Contact Us

At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

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Email: info@quectel.com

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https://www.quectel.com/contact/.

For technical support, or to report documentation errors, please visit:
https://www.quectel.com/tech-support/.
Or email us at: support@quectel.com.

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Revision History

Version	Date	Author	Note
-	2026-08-10	Simon Sheng/ Toby Wang/ Piers Ji/ Rainey Liao	Creation of the document
1.0	2026-08-10	Simon Sheng/ Toby Wang/ Piers Ji/ Rainey Liao	First official release



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